

CropMatch Analog Engine — parcel brief

Chaco Paraguayo · centroid -22.5000, -60.0000 · 12.5 km2 · generated 2026-09-12T09:59:22+00:00 · mode offline · result 6evrfxv_c7a8cd50_9ff48e

- Similarity is a percentile rank against the reference grid, not a probability of success or an accuracy.
- Technique effects are summarised from the cited sources; constraints are screening heuristics. Review with a local agronomist before acting.
- Values marked 'not available' were not retrieved or not computable; nothing has been imputed.

1. Biophysical profile

24 of 24 dimensions available. z = standard deviations from the reference-grid mean. Land cover: Grassland (ESA WorldCover 2021). Dry quarter (months): [6, 7, 8].

Group	Dimension	Value	z	Source
climate	Mean annual temperature	24.7 °C	+0.80	WorldClim 2.1 BIO1
climate	Temperature seasonality (SD x 100)	361 °Cx100	-0.58	WorldClim 2.1 BIO4
climate	Max temperature of warmest month	35.6 °C	+0.62	WorldClim 2.1 BIO5
climate	Min temperature of coldest month	11.8 °C	+0.71	WorldClim 2.1 BIO6
climate	Annual precipitation	685 mm	+0.09	WorldClim 2.1 BIO12
climate	Precipitation seasonality (CV)	55.3 %	-0.60	WorldClim 2.1 BIO15
climate	Aridity index (P / PET)	0.368	-0.22	WorldClim BIO12 / Hargreaves PET from WorldClim monthly Tmin, Tmax
water	Mean annual evapotranspiration	407 mm/yr	-0.15	MODIS MOD16A3GF.061 ET_500m
water	Water use efficiency (GPP / ET)	1.84 g C / kg H2O	-0.01	MODIS MOD17A3HGF.061 GPP / MOD16A3GF.061 ET
water	NDWI, dry-season mean	-0.492	-0.65	Sentinel-2 L2A (B03, B08)
water	NDMI, dry-season mean	-0.010	-0.13	Sentinel-2 L2A (B08, B11)
water	Sentinel-1 VV backscatter, dry-season mean	-9.51 dB	+0.29	Sentinel-1 IW GRD -> RTC gamma0 VV
soil	pH (H2O), 0–30 cm	6.88 pH	+0.01	ISRIC SoilGrids v2.0 phh2o
soil	Soil organic carbon, 0–30 cm	16.0 g/kg	-0.31	ISRIC SoilGrids v2.0 soc
soil	Clay, 0–30 cm	32.1 %	+0.94	ISRIC SoilGrids v2.0 clay
soil	Sand, 0–30 cm	33.5 %	-0.99	ISRIC SoilGrids v2.0 sand
soil	Cation exchange capacity, 0–30 cm	18.9 cmol(c)/kg	-0.23	ISRIC SoilGrids v2.0 cec
soil	Bulk density, 0–30 cm	1.35 kg/dm³	+0.14	ISRIC SoilGrids v2.0 bdod
soil	Coarse fragments, 0–30 cm	3.05 vol %	-1.52	ISRIC SoilGrids v2.0 cfvo
terrain	Elevation	133 m	-0.64	Copernicus DEM GLO-90
terrain	Mean slope	0.271 °	-0.75	Copernicus DEM GLO-90
terrain	Insolation index (equinox noon, flat = 1)	1.000	+0.49	Copernicus DEM GLO-90
vegetation	NDVI amplitude (peak - trough)	0.408	+0.75	MODIS MOD13Q1.061
vegetation	Growing season length	208 days	+0.62	MODIS MOD13Q1.061

2. Closest analogs in documented innovation regions

Compared with 1981 of 2000 reference cells (19 excluded for sharing too few dimensions). Effective group weights: climate 0.25, water 0.30, soil 0.25, terrain 0.10, vegetation 0.10

#	Region	Cell	Percentile	Distance	Closest dimensions	Largest divergences
1	Sonora (Costa de Hermosillo, Yaqui and Mayo valleys) (Mexico)	27.625, -110.125	99.5	0.690	NDMI, dry-season mean; Mean annual temperature; Cation exchange capacity, 0–30 cm	Precipitation seasonality (CV); pH (H2O), 0–30 cm; Max temperature of warmest month
2	Western Rajasthan (India)	28.875, 71.125	99.4	0.704	Mean slope; Insolation index (equinox noon, flat = 1); Water use efficiency (GPP / ET)	Temperature seasonality (SD x 100); NDWI, dry-season mean; Sand, 0–30 cm
3	San Juan (Tulum, Ullum-Zonda valleys) (Argentina)	-31.875, -68.875	99.1	0.724	Precipitation seasonality (CV); Water use efficiency (GPP / ET); Cation exchange capacity, 0–30 cm	Mean annual temperature; Coarse fragments, 0–30 cm; NDWI, dry-season mean
4	Mendoza oases (Argentina)	-33.125, -69.125	99.0	0.725	Cation exchange capacity, 0–30 cm; Sentinel-1 VV backscatter, dry-season mean; Soil organic carbon, 0–30 cm	Mean annual temperature; Sand, 0–30 cm; Growing season length
5	Xinjiang oases (Tarim and Junggar margins) (China)	39.625, 76.875	98.7	0.741	Mean slope; Insolation index (equinox noon, flat = 1); NDMI, dry-season mean	Temperature seasonality (SD x 100); Elevation; Growing season length

3. Transferable techniques (ranked)

Transferability = analog similarity (source-region percentile / 100) x share of the technique's constraints verified as met by this parcel.

1.00 — Bed planting of irrigated wheat (furrow-irrigated raised beds)

Sonora (Costa de Hermosillo, Yaqui and Mayo valleys) · irrigation · capital low · crops: wheat

Documented effect: Trial of straw management and nitrogen rates in a bed-planting system reporting wheat and maize yield responses.

Evidence: Limon-Ortega, A., Sayre, K.D. & Francis, C.A. (2000). Wheat and maize yields in response to straw management and nitrogen under a bed planting system. *Agronomy Journal* 92: 295–302. <https://doi.org/10.2134/agronj2000.922295x> [doi verified]

Adaptation: Row spacing and bed width must match available machinery.

1.00 — Permanent raised beds with crop-residue retention under furrow irrigation

Sonora (Costa de Hermosillo, Yaqui and Mayo valleys) · mulching cover cropping · capital medium · crops: wheat, maize, soybean

Documented effect: Long-term trial in the Yaqui Valley comparing tillage and residue management on beds; reports effects on soil quality indicators.

Evidence: Verhulst, N. et al. (2011). Soil quality as affected by tillage-residue management in a wheat-maize irrigated bed planting system. *Plant and Soil* 340: 453–466. <https://doi.org/10.1007/s11104-010-0618-5> [doi verified]

Adaptation: Requires bed-shaping and residue-tolerant planters; residue must be managed so it doesn't block furrows.

1.00 — Optical-sensor (NDVI) based nitrogen top-dressing in irrigated wheat

Sonora (Costa de Hermosillo, Yaqui and Mayo valleys) · low cost sensing · capital low · crops: wheat, maize

Documented effect: Reports reduced nitrogen use and improved farm income in irrigated spring wheat in the Yaqui Valley using sensor-based nitrogen management.

Evidence: Ortiz-Monasterio, J.I. & Raun, W. (2007). Reduced nitrogen and improved farm income for irrigated spring wheat in the Yaqui Valley, Mexico, using sensor based nitrogen management. *Journal of Agricultural Science* 145: 215–222. <https://doi.org/10.1017/S0021859607006995> [doi verified metadata]

Adaptation: Needs an N-rich reference strip in each field; algorithms are calibrated per crop and region.

0.99 — Prosopis cineraria (khejri) parkland agroforestry with arable crops

Western Rajasthan · agroforestry · capital low · crops: pearl millet, cluster bean, moth bean, fodder

Documented effect: Studies how tree density affects the productivity of a Prosopis cineraria agroforestry system in north-western India.

Evidence: Singh, G., Mutha, S. & Bala, N. (2007). Effect of tree density on productivity of a Prosopis cineraria agroforestry system in North Western India. *Journal of Arid Environments* 70: 152–163. <https://doi.org/10.1016/j.jaridenv.2006.12.003> [doi verified]

Adaptation: Prosopis cineraria is native to the Thar; do not substitute invasive Prosopis juliflora. Lopping regime controls competition with crops.

0.99 — Pearl millet – legume (cluster bean, moth bean) rotation and intercropping

Western Rajasthan · rotation · capital low · crops: pearl millet, cluster bean, moth bean, cowpea

Documented effect: not available (no verified source linked for this entry)

Evidence: ICAR-Central Arid Zone Research Institute (CAZRI), Jodhpur — arid cropping-systems research. <https://www.cazri.res.in/> [url ok]

Adaptation: Legume choice must match season length; guar (cluster bean) doubles as a cash crop.

0.99 — Tanka / kund: covered underground cisterns fed by a prepared catchment

Western Rajasthan · water harvesting · capital low · crops: household water, livestock, kitchen garden

Documented effect: Survey of India's traditional water-harvesting systems, including the tankas and kunds of the Thar desert.

Evidence: Agarwal, A. & Narain, S. (eds.) (1997). Dying Wisdom: Rise, Fall and Potential of India's Traditional Water Harvesting Systems. Centre for Science and Environment, New Delhi. not available [no link]

Adaptation: Catchment surface must be kept clean; lining quality determines losses.

0.99 — Tree windbreaks (e.g. poplar rows) around irrigated plots

San Juan (Tulum, Ullum-Zonda valleys) · windbreaks · capital low · crops: wine grape, vegetables, fruit trees

Documented effect: Reviews windbreak research in North American agriculture: reduced wind speed in the sheltered zone, microclimate changes and crop and soil-erosion effects.

Evidence: Brandle, J.R., Hodges, L. & Zhou, X.H. (2004). Windbreaks in North American agricultural systems. *Agroforestry Systems* 61: 65–78. <https://doi.org/10.1023/B:AGFO.0000028990.31801.62> [doi verified]

Adaptation: Evidence is from North America; windbreak trees need water themselves in arid areas and shade the rows next to them.

0.99 — Inter-row cover crops in irrigated vineyards

Mendoza oases · mulching cover cropping · capital low · crops: wine grape

Documented effect: not available (no verified source linked for this entry)

Evidence: INTA EEA Mendoza extension material on vineyard cover crops (no specific trial report linked). not available [no link]

Adaptation: Choose species and mowing dates so cover-crop water use does not coincide with critical vine stages.

0.99 — Alternate partial root-zone irrigation (PRD)

Xinjiang oases (Tarim and Junggar margins) · irrigation · capital low · crops: maize, cotton, grape, pear

Documented effect: Reviews the physiology of controlled alternate partial root-zone irrigation and its effect on water-use efficiency.

Evidence: Kang, S. & Zhang, J. (2004). Controlled alternate partial root-zone irrigation: its physiological consequences and impact on water use efficiency. *Journal of Experimental Botany* 55: 2437–2446. <https://doi.org/10.1093/jxb/erh249> [doi verified]

Adaptation: Needs alternating furrows or dual drip lines and disciplined switching; the benefit is small on shallow soils.

0.99 — Oasis farmland shelterbelt networks

Xinjiang oases (Tarim and Junggar margins) · windbreaks · capital low · crops: cotton, wheat, fruit

Documented effect: Reviews windbreak research: reduced wind speed in the sheltered zone, microclimate changes and effects on crops and erosion.

Evidence: Brandle, J.R., Hodges, L. & Zhou, X.H. (2004). Windbreaks in North American agricultural systems. *Agroforestry Systems* 61: 65–78. <https://doi.org/10.1023/B:AGFO.0000028990.31801.62> [doi verified]

Adaptation: The cited review is North American; shelterbelt trees need an irrigation allocation in oases.

0.98 — Subsidised drip and sprinkler irrigation for smallholder cash crops

Gujarat (Kutch, Saurashtra) · irrigation · capital medium · crops: cotton, groundnut, banana, vegetables

Documented effect: Empirical study of the economics, adoption determinants and impacts of micro-irrigation technologies in India.

Evidence: Namara, R.E., Nagar, R.K. & Upadhyay, B. (2007). Economics, adoption determinants, and impacts of micro-irrigation technologies: empirical results from India. *Irrigation Science* 25: 283–297. <https://doi.org/10.1007/s00271-007-0065-0> [doi verified]

Adaptation: Adoption in the source region was supported by a state subsidy scheme; without one, capital cost is the main barrier.

0.97 — Rainwater cisterns for household and small-plot production (P1MC / P1+2 programmes)

Petrolina–Juazeiro (São Francisco valley) · water harvesting · capital low · crops: vegetables, fruit trees, small livestock

Documented effect: Viability assessment of the One Million Cisterns Programme as community rainwater management in Brazil's semiarid region.

Evidence: Cardoso Castro, P.P., Vachkova, M., Ravena, N. & Veloso, N. (2024). The One Million Cisterns Programme — a viability assessment of community rainwater management in Brazil. *Frontiers in Sustainability* 5. <https://doi.org/10.3389/frsus.2024.1401440> [doi verified]

Adaptation: Storage volume must be sized to dry-season length and use; first-flush diversion protects water quality.

0.97 — Drip and micro-sprinkler irrigated tropical fruit (mango, grape) with scheduled irrigation cut-off

Petrolina–Juazeiro (São Francisco valley) · irrigation · capital high · crops: mango, table grape, guava

Documented effect: not available (no verified source linked for this entry)

Evidence: Embrapa Semiárido irrigated-fruit research (no specific trial report linked). not available [no link]

Adaptation: Depends on a reliable perennial water source (the São Francisco river in the source region) and export market access.

0.97 — Conservation farming: ripping/sub-soiling with mulch retention in rainfed maize

Kenya ASAL (Machakos, Makueni, Kitui, Laikipia) · mulching cover cropping · capital low · crops: maize, sorghum, beans

Documented effect: On-farm action research in East and Southern Africa reporting yields and rain-water productivity under conservation farming compared with conventional tillage.

Evidence: Rockström, J. et al. (2009). Conservation farming strategies in East and Southern Africa: Yields and rain water productivity from on-farm action research. *Soil & Tillage Research* 103: 23–32. <https://doi.org/10.1016/j.still.2008.09.013> [doi verified]

Adaptation: Mulch competes with livestock feed; rippers need draught power or a service provider.

0.97 — Maize – pigeonpea intercropping

Kenya ASAL (Machakos, Makueni, Kitui, Laikipia) · rotation · capital low · crops: maize, pigeonpea

Documented effect: Study of nitrogen use in maize–pigeonpea intercrops under semi-arid conditions in Kenya.

Evidence: Wanderi, S.W., Mburu, M.W.K., Silim, S.N. & Kihanda, F.M. (2011). Nitrogen Use in Maize (Zea mays)–Pigeonpea (Cajanus cajan) Intercrop in Semi-arid Conditions of Kenya. In: *Innovations as Key to the Green Revolution in Africa*, pp. 563–570. Springer. https://doi.org/10.1007/978-90-481-2543-2_58 [doi verified]

Adaptation: Choose pigeonpea maturity group to match rainfall season length.

4. Blocked techniques

Technique	Source region	Why it does not transfer
Khadir: runoff from a rocky catchment impounded behind an earthen bund, then cropped on residual moisture	Western Rajasthan	Requires Aridity index (P / PET) <= 0.3; this parcel: 0.37; Requires Mean slope >= 0.3 °; this parcel: 0.27 °
Shelterbelts of drought-hardy trees against wind erosion and sand movement	Western Rajasthan	Requires Sand, 0–30 cm >= 40 %; this parcel: 33 %
Conversion of furrow-irrigated vineyards to pressurised drip	San Juan (Tulum, Ullum-Zonda valleys)	Requires Aridity index (P / PET) <= 0.3; this parcel: 0.37
Controlled deficit irrigation in red wine grapes	Mendoza oases	Requires Min temperature of coldest month <= 8 °C; this parcel: 12 °C
Anti-hail netting over vineyards and orchards	Mendoza oases	Requires Min temperature of coldest month <= 8 °C; this parcel: 12 °C; Requires Mean annual temperature <= 20 °C; this parcel: 25 °C
Drip irrigation under plastic film mulch for cotton	Xinjiang oases (Tarim and Junggar margins)	Requires Aridity index (P / PET) <= 0.3; this parcel: 0.37; Requires Min temperature of coldest month <= 5 °C; this parcel: 12 °C
Pre-season (winter or spring) flood leaching of accumulated salts	Xinjiang oases (Tarim and Junggar margins)	Requires Aridity index (P / PET) <= 0.3; this parcel: 0.37
Community check dams and well recharge (Saurashtra recharge movement)	Gujarat (Kutch, Saurashtra)	Requires Precipitation seasonality (CV) >= 80 %; this parcel: 55 %
Tidal regulators, spreading channels and recharge to halt seawater ingress into coastal aquifers	Gujarat (Kutch, Saurashtra)	Requires Elevation <= 100 m; this parcel: 133 m
Underground (subsurface) dams across alluvial valley floors	Petrolina–Juazeiro (São Francisco valley)	Requires Precipitation seasonality (CV) >= 60 %; this parcel: 55 %
Fanya juu terraces (soil thrown uphill to build bench terraces over time)	Kenya ASAL (Machakos, Makueni, Kitui, Laikipia)	Requires Mean slope >= 1 °; this parcel: 0.27 °
Sand dams across seasonal sandy riverbeds	Kenya ASAL (Machakos, Makueni, Kitui, Laikipia)	Requires Precipitation seasonality (CV) >= 60 %; this parcel: 55 %
Fertilizer micro-dosing (small doses placed in the planting hole)	West African Sahel (Yatenga, Maradi, Tahoua, Zinder)	Requires Soil organic carbon, 0–30 cm <= 15 g/kg; this parcel: 16 g/kg
Pearl millet – cowpea / groundnut rotation on sandy soils	West African Sahel (Yatenga, Maradi, Tahoua, Zinder)	Requires Sand, 0–30 cm >= 40 %; this parcel: 33 %
Contour stone bunds (cordons pierreux)	West African Sahel (Yatenga, Maradi, Tahoua, Zinder)	Requires Mean slope >= 0.3 °; this parcel: 0.27 °
Tree windbreaks against wind erosion on sandy Sahelian fields (e.g. Majjia Valley, Niger)	West African Sahel (Yatenga, Maradi, Tahoua, Zinder)	Requires Sand, 0–30 cm >= 50 %; this parcel: 33 %
Area enclosures: degraded hillsides closed to grazing and cutting for regeneration	Tigray highlands	Requires Mean slope >= 3 °; this parcel: 0.27 °
Stone bunds, check dams and percolation trenches at catchment scale	Tigray highlands	Requires Mean slope >= 1 °; this parcel: 0.27 °
On-farm winter flood recharge (Flood-MAR) of groundwater on dormant fields	Central Valley	Requires Precipitation seasonality (CV) >= 60 %; this parcel: 55 %
Gypsum application to improve infiltration in sodic or low-salinity-water-sealed soils	Central Valley	Requires pH (H ₂ O), 0–30 cm between 7 and 9.5; this parcel: 6.88
Limans: small runoff-fed tree groves in wadi depressions	Negev Desert	Requires Aridity index (P / PET) <= 0.3; this parcel: 0.37; Requires Mean slope >= 0.3 °; this parcel: 0.27 °
Runoff agriculture with microcatchments and contour-bounded runoff plots	Negev Desert	Requires Aridity index (P / PET) <= 0.35; this parcel: 0.37; Requires Mean slope >= 0.5 °; this parcel: 0.27 °
Break-crop rotations (canola, pulses) in wheat sequences	Mallee (Victoria / South Australia)	Requires Min temperature of coldest month <= 10 °C; this parcel: 12 °C
Clay spreading and delving on water-repellent sandy soils	Mallee (Victoria / South Australia)	Requires Sand, 0–30 cm >= 70 %; this parcel: 33 %
Fog collectors (atrapanieblas) for water supply on coastal hills	Norte Chico (Copiapó, Huasco, Elqui, Limarí)	Requires Aridity index (P / PET) <= 0.2; this parcel: 0.37
Drip irrigation of avocado and table grape planted on steep hillsides	Norte Chico (Copiapó, Huasco, Elqui, Limarí)	Requires Mean slope >= 2 °; this parcel: 0.27 °
Quesungual slash-and-mulch agroforestry (no burning, pruned trees, permanent mulch)	Lempira (Quesungual agroforestry)	Requires Annual precipitation >= 800 mm; this parcel: 685 mm; Requires Mean slope >= 2 °; this parcel: 0.27 °

5. Data sources and provenance

- Mean annual ET and WUE come from MODIS MOD16A3GF.061 / MOD17A3HGF.061, not ECOSTRESS: ECOSTRESS (ECO3ETPTJPL, ECO4WUE) is Tier 2 and needs an Earthdata token. Values are real MODIS granules, see provenance.
- Sentinel-2 L2A is read from Element84 Earth Search (AWS Open Data) and Sentinel-1 RTC and MODIS from Microsoft Planetary Computer, which need no credentials. They serve the same ESA / NASA products as Copernicus Data Space and AppEEARS; product IDs in the audit log are the original ESA / NASA IDs.
- Aridity index uses Hargreaves PET computed from WorldClim monthly temperatures, not Penman–Monteith.
- GEDI, AIRS and ECOSTRESS (Tier 2) and Landsat / VIIRS long-term series are not part of this vector yet.

Source	Dimensions	Status	Note
WorldClim v2.1	mat, temp_seasonality, tmax_warmest, tmin_coldest, annual_precip, precip_seasonality, aridity_index	ok	
ISRIC SoilGrids v2.0	ph, soc, clay, sand, cec, bdod, cfvo	ok	
Copernicus DEM GLO-90	elevation, slope, insolation	ok	
MODIS MOD13Q1 phenology	ndvi_amplitude, growing_season_days	ok	
MODIS MOD16A3GF/MOD17A3HGF	et_annual, wue	ok	
Sentinel-2 L2A	ndwi_dry, ndmi_dry	ok	
Sentinel-1 RTC	s1_vv_dry	ok	

90 logged reads: NASA LP DAAC modis-13Q1-061 via Planetary Computer (61); Microsoft Planetary Computer STAC (5); NASA LP DAAC modis-16A3GF-061 via Planetary Computer (4); Sentinel-2 L2A via Element84 Earth Search (AWS Open Data) (4); Sentinel-1 IW GRD -> RTC via Planetary Computer (4); NASA LP DAAC modis-17A3HGF-061 via Planetary Computer (4); Element84 Earth Search STAC (2); Copernicus DEM GLO-90 (AWS Open Data) (2); ESA WorldCover via Planetary Computer (2); WorldClim v2.1 (local GeoTIFF, geodata.ucdavis.edu) (1); ISRIC SoilGrids v2.0 REST API (1)

Satellite scenes used

- S2B_20KRA_20240607_0_L2A · 2024-06-07 · L2A surface reflectance, processing baseline 05.10
- S2B_20KRA_20250722_0_L2A · 2025-07-22 · L2A surface reflectance, processing baseline 05.11
- S2C_20KQA_20250816_0_L2A · 2025-08-16 · L2A surface reflectance, processing baseline 05.11
- S2B_20KRA_20240826_0_L2A · 2024-08-26 · L2A surface reflectance, processing baseline 05.11
- S1C_IW_GRDH_1SDV_20250805T092727_20250805T092752_003535_007159_F162 · 2025-08-05 · L1 GRD, radiometric terrain correction (gamma0), 10 m
- S1C_IW_GRDH_1SDV_20250630T092725_20250630T092750_003010_006215_5D7E · 2025-06-30 · L1 GRD, radiometric terrain correction (gamma0), 10 m
- S1C_IW_GRDH_1SDV_20250606T092723_20250606T092748_002660_0057FC_C33D · 2025-06-06 · L1 GRD, radiometric terrain correction (gamma0), 10 m
- S1C_IW_GRDH_1SDV_20250829T092716_20250829T092741_003885_007BDB_3847 · 2025-08-29 · L1 GRD, radiometric terrain correction (gamma0), 10 m